

ProtoCoach-Yoga: Explainable Yoga Coaching via Pose Relationship Prototype Learning and Small Language Model Feedback

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1. Introduction

Human activity recognition (HAR) has achieved significant progress in analyzing and recognizing human movements from sensor- and vision-based data through advances in deep learning and skeleton-based representation learning [1], [2]. These developments have enabled intelligent applications in healthcare, rehabilitation, sports coaching, and fitness. In particular, recent studies have demonstrated promising performance in yoga pose recognition using RGB images, pose estimation techniques, and skeleton-based analysis [3]–[5], [14]. However, most existing approaches primarily focus on pose classification, producing activity labels with limited support for posture assessment or corrective guidance.

For practical yoga coaching, recognizing a pose is only the first step. Effective coaching requires understanding posture quality, body alignment, and movement execution to provide meaningful corrective feedback. Recent explainable artificial intelligence (XAI) methods have improved the transparency of human activity recognition models through concept-based reasoning [6], visual explanation techniques [8], feature attribution [9], and prototype-based explanations [10]. Although these methods provide insights into model predictions, they generally explain *why* a prediction is made rather than *how* a posture should be improved. Consequently, they provide limited support for interactive coaching scenarios where users require personalized and actionable guidance.

To address this limitation, this paper proposes ProtoCoach-Yoga, an explainable yoga coaching framework that integrates pose relationship prototype learning with Small Language Model (SLM)-based feedback generation.

The proposed framework first converts skeleton sequences into pose relationship representations that describe body posture and alignment using interpretable features such as joint angles, limb orientation, body symmetry, and relative joint distances. Similar pose relationship representations are then clustered to learn representative pose relationship prototypes, where each prototype summarizes a recurring body alignment pattern shared across multiple posture samples. For a new input posture, the framework retrieves the corresponding pose relationship prototype and compares the input pose relationship representation with the retrieved prototype to identify body

alignment differences. These differences are then provided as structured input to a Small Language Model (SLM) for generating personalized coaching feedback.

To investigate the feasibility of explainable prototype-based yoga coaching, this paper addresses the following research questions:

- **RQ1:** Can interpretable pose relationship representations be constructed from skeleton motion data to support prototype learning?
- **RQ2:** Can representative pose relationship prototypes provide an effective intermediate reasoning layer between skeleton-based motion analysis and SLM-based coaching feedback?
- **RQ3:** Can prototype-grounded posture reasoning enable transparent and explainable AI-assisted coaching?

The remainder of this paper is organized as follows. Section 2 reviews related work on yoga pose recognition, explainable human activity recognition, and prototype-based learning, and identifies the research gap addressed by this work. Section 3 presents the proposed ProtoCoach-Yoga framework, including body relationship representation, pose relationship prototype learning, prototype-based posture assessment, and SLM-based coaching feedback generation. Section 4 presents the experimental design and preliminary results, including relationship feature analysis, prototype space visualization, and an explainable prototype-based coaching example. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related Work

Recent advances in human activity recognition (HAR), explainable artificial intelligence (XAI), and prototype-based learning have enabled intelligent systems for movement analysis and human-centered assistance. This section reviews representative studies on yoga pose recognition and explainable HAR and identifies the research gap addressed by ProtoCoach-Yoga.

2.1 Yoga Pose Recognition

Skeleton-based representation learning has become a dominant approach for human activity recognition because it effectively models human body motion while remaining relatively robust to appearance variations. ST-GCN [1] first represented the human skeleton as a spatial-temporal graph for action recognition, while 2s-AGCN [2] further improved recognition performance through adaptive graph learning and a two-stream architecture.

For yoga analysis, Verma *et al.* [3] introduced Yoga-82, providing a benchmark dataset for fine-grained yoga pose recognition. BlazePose [4] and OpenPose [5] enabled reliable human pose estimation from RGB images, making practical skeleton-based posture analysis possible. More recently, Kou and Li [14] demonstrated that deep learning with attention mechanisms can further improve yoga pose recognition accuracy;

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Table 1. Comparison of representative studies and ProtoCoach-Yoga

Method	Skeleton	Explainable	Prototype/Concept Learning	Coaching Feedback
ST-GCN [1]	✓	✗	✗	✗
2s-AGCN [2]	✓	✗	✗	✗
Yoga-82 [3]	✓	✗	✗	✗
X-CHAR [6]	✓	✓	✓	✗
ProtoPNet [10]	✗	✓	✓	✗
PCBEAR [12]	✓	✓	✓	✗
ProtoCoach-Yoga	✓	✓	✓	✓

however, their primary objective is pose classification rather than posture assessment or personalized coaching.

2.2 Explainable Human Activity Recognition

To improve the transparency of deep learning models, various explainability techniques have been proposed for human activity recognition. X-CHAR [6] introduced concept-based explanations for complex activity recognition by representing activities through human-interpretable concepts, while Huang *et al.* [7] proposed an explainable deep learning framework that combines activity recognition with visual explanation mechanisms. More general explainability techniques, including Grad-CAM [8] and TCAV [9], provide visual and concept-based interpretations of model predictions, respectively.

Recently, prototype- and concept-based learning have emerged as effective approaches for interpretable deep learning. ProtoPNet [10] introduced prototype reasoning through representative examples, and Label-Free Concept Bottleneck Models [11] demonstrated that interpretable concepts can be learned without manually annotated concept labels. More recently, PCBEAR [12] proposed pose concept bottlenecks to explain skeleton-based action recognition through representative pose concepts. While these methods improve model interpretability, they primarily focus on explaining recognition decisions rather than assessing posture quality or generating personalized coaching feedback.

2.3 Research Gap

Although previous studies have significantly advanced yoga pose recognition, explainable HAR, and prototype-based learning, several limitations remain. Existing yoga recognition systems primarily emphasize accurate pose classification without explicitly assessing posture quality or body alignment. Likewise, existing explainability methods mainly interpret model predictions instead of providing actionable coaching guidance.

Furthermore, recent prototype- and concept-based approaches, including ProtoPNet [10], Label-Free Concept Bottleneck Models [11], and PCBEAR [12], have demonstrated that representative concepts can improve model interpretability. However, these methods primarily explain recognition decisions rather than generating personalized coaching feedback grounded in interpretable posture concepts or connecting these concepts to natural-language coaching and posture correction.

Table 1 compares representative studies with the proposed ProtoCoach-Yoga framework. As shown, existing methods mainly address activity recognition or model interpretability independently, whereas ProtoCoach-Yoga integrates interpretable pose relationship prototype learning with SLM-based coaching feedback for transparent posture assessment and personalized coaching.

Unlike existing explainability methods that primarily interpret recognition decisions, ProtoCoach-Yoga introduces pose relationship prototypes as an intermediate reasoning layer between posture assessment and corrective coaching feedback. By comparing the input pose relationship representation with the retrieved pose relationship prototype, the framework identifies interpretable body alignment differences and provides these differences as structured input to a Small Language Model (SLM), enabling personalized corrective coaching rather than merely explaining recognition results.

3. ProtoCoach-Yoga Framework

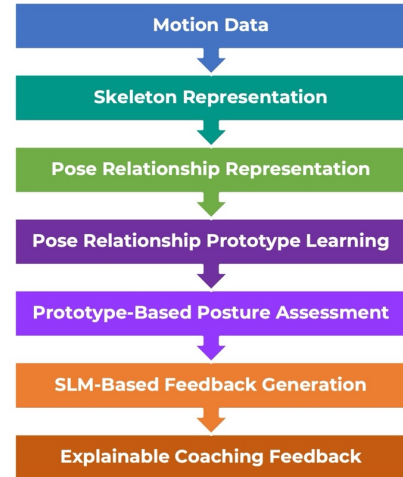


Figure 1. Overview of the proposed ProtoCoach-Yoga framework.

This section presents the proposed ProtoCoach-Yoga framework for explainable yoga coaching. Unlike conventional yoga coaching systems that primarily map motion data to pose labels, ProtoCoach-Yoga introduces pose relationship prototypes as an intermediate reasoning layer between posture analysis and

language generation. A pose relationship prototype is a representative body alignment pattern learned from similar pose relationship representations and serves as an interpretable reference for posture assessment. By combining pose relationship prototype learning with Small Language Model (SLM)-based feedback generation, the proposed framework bridges skeleton-based motion analysis and human-understandable coaching feedback. A Small Language Model (SLM) is adopted because the structured posture differences extracted from prototype comparison require lightweight reasoning rather than broad world knowledge. Compared with larger language models, an SLM offers lower computational cost and faster inference while remaining sufficient for generating personalized coaching feedback.

As illustrated in Figure 1, ProtoCoach-Yoga transforms motion data into explainable coaching feedback through a sequence of interpretable processing stages. Together, these stages establish a transparent pipeline for posture assessment and personalized coaching, providing a foundation for explainable AI-assisted yoga coaching.

3.1 Framework Overview

Figure 1 illustrates the overall workflow of the proposed ProtoCoach-Yoga framework. Starting from motion data, the framework first constructs a skeleton representation and derives an interpretable pose relationship representation describing the observed posture. Representative pose relationship prototypes are subsequently learned by clustering similar pose relationship representations, where each cluster captures a recurring body alignment pattern. Each prototype serves as an interpretable reference for posture assessment. For a new input posture, the framework performs prototype-based posture assessment by comparing its pose relationship representation with the learned prototypes to identify body alignment characteristics and posture deviations. Finally, the retrieved pose relationship prototype together with the identified body alignment differences is supplied to a Small Language Model (SLM) [13], which generates personalized and interpretable coaching feedback grounded in this structured posture information. The resulting explainable coaching feedback provides users with human-understandable guidance for improving posture quality and body alignment.

3.2 Pose Relationship Prototype Learning

The first stage of ProtoCoach-Yoga aims to learn representative pose relationship prototypes from interpretable pose relationship representations. Rather than relying directly on raw joint coordinates or latent feature embeddings, the framework represents each posture using body relationship features that describe body geometry and alignment. These representations characterize posture through meaningful body relationships, including joint configurations, body alignment, body symmetry, and relative joint distances.

Representative pose relationship prototypes are learned by applying K-Means clustering to standardized body relationship feature vectors. Each cluster centroid represents a recurring body

alignment pattern shared across multiple posture samples and serves as an interpretable prototype. Unlike pose labels, these prototypes capture common body relationship patterns that may be shared across different yoga postures, providing an intermediate semantic representation for posture assessment.

Table 2. Example of a learned prototype (Concept 1) represented by representative body relationship feature values.

Relationship Feature	Representative Value
Left knee angle	130.65
Left thigh–trunk angle	105.10
Right thigh–trunk angle	111.17
Foot distance	49.05
Knee angle difference	34.12

Table 2 presents an example of a learned pose relationship prototype. Each prototype stores representative values for the extracted body relationship features, including joint angles, body alignment, and relative body distances. These representative feature values summarize the characteristic posture pattern of the corresponding cluster and serve as interpretable references for subsequent posture assessment.

Compared with raw joint coordinates, pose relationship representations explicitly describe body geometry and alignment through relative body relationships. Consequently, prototype learning is performed in a feature space that preserves semantic posture information, enabling representative body alignment patterns to be learned in an interpretable manner.

3.3 Prototype-Based Posture Assessment

Given a new input posture, the framework first extracts the same 33-dimensional body relationship feature representation used during prototype learning. The extracted feature vector is standardized using the previously learned feature normalization and projected into the learned prototype space.

Table 3. Example of Prototype-Based Posture Assessment

Stage	Example Output
Input clip	Y007_00150
Predicted pose	Upward Facing Dog (66.3%)
Retrieved prototype	Concept 1
Top body relationship differences	Left knee angle ↓ 35.44
	Left thigh–trunk angle ↓ 32.10
	Knee angle asymmetry ↑ 28.37
	Right thigh–trunk angle ↑ 26.23
	Left hand–foot distance ↓ 20.59

Prototype-based posture assessment is then performed by comparing the standardized input feature vector with all learned prototype centroids. The nearest prototype is identified using the Euclidean distance between the input feature vector and each prototype centroid. The retrieved prototype represents the most

similar body relationship pattern and serves as an interpretable reference for posture assessment.

Table 3 presents an example of prototype-based posture assessment. For the input posture, the framework predicts the yoga pose, retrieves the nearest prototype, and compares the representative body relationship features of the input posture with those of the retrieved prototype. The resulting feature differences describe explicit posture deviations, including differences in joint angles, body alignment, body symmetry, and relative body distances.

Unlike conventional pose classification, which primarily predicts activity labels, prototype-based posture assessment explains how the observed posture differs from a representative body alignment pattern. These interpretable body relationship differences provide transparent evidence for posture assessment and establish the structured information required for subsequent SLM-based coaching feedback generation.

3.4 SLM-Based Coaching Feedback

The final stage transforms prototype-based posture assessment into natural-language coaching feedback. Rather than relying solely on raw skeleton data, the framework uses the body alignment differences identified through comparison with the retrieved prototype as structured input to a Small Language Model (SLM) [13]. The extracted posture differences are organized into a structured prompt containing the predicted pose, interpreted posture issues, and task instructions before being provided to the SLM. Figure 2 illustrates an example of this process, where prototype-based posture differences are converted into personalized coaching recommendations through SLM-based feedback generation.

SLM Prompt Example	
Input posture:	Upward Facing Dog
Detected posture issues:	• Lower-body extension differs from the learned prototype • Hip and thigh extension differs from the learned prototype • Lower-body symmetry differs from the learned prototype • Hand-foot relationship differs from the learned prototype
Instruction:	Generate exactly three short yoga coaching instructions. Do not mention numerical values. Use safe and supportive language.

Figure 2. Example of a structured SLM prompt for prototype-based coaching feedback generation.

Figure 2 presents an example of the structured SLM prompt generated from prototype-based posture assessment. The prompt contains the predicted pose, interpreted posture issues derived from comparison with the retrieved prototype, and task instructions that guide the SLM to generate personalized coaching feedback.

By using prototype-based posture assessment as an intermediate reasoning layer, the generated feedback remains

explicitly connected to interpretable body alignment differences rather than opaque model predictions. Consequently, ProtoCoach-Yoga provides a transparent and explainable coaching framework that can be extended to broader movement-learning applications, including rehabilitation, elderly exercise support, sports coaching, and home fitness assistance.

4. Preliminary Results

This section presents preliminary results to demonstrate the feasibility of the proposed ProtoCoach-Yoga framework. Rather than providing a comprehensive quantitative evaluation, the presented analyses illustrate how skeleton-based relationship features can be organized into interpretable pose relationship representations that support prototype learning and explainable coaching. These preliminary findings provide initial evidence for the proposed framework and motivate future investigations on prototype-driven posture assessment and SLM-based feedback generation.

4.1 Experimental Design

The proposed ProtoCoach-Yoga framework was implemented using the hardware and software environment summarized in Table 4.

Table 4. Experimental Setup

Component	Details
Processor	Intel Core i7-13700
GPU	NVIDIA GeForce RTX 4080
RAM	64 GB
Python	3.11.15
PyTorch	2.5.1
CUDA	12.1

The preliminary evaluation was conducted using the proposed DSZ-Yoga dataset, which was collected using the Sony mocopi motion capture system equipped with 12 wearable sensors. The dataset consists of skeleton motion sequences collected from 12 participants performing nine yoga postures. A summary of the dataset characteristics is presented in Table 5.

Table 5. Dataset Summary

Attribute	Specification
Dataset	DSZ-Yoga
Motion Capture	Sony mocopi Pro Kit
Participants	12
Yoga Postures	9
Motion Clips	2,071
Input Modality	Skeleton motion sequences
Feature Representation	Pose relationship features

Each motion clip consists of a sequence of 3D skeleton joint coordinates captured by the Sony mocopi motion capture system. Rather than using raw joint coordinates directly, the proposed framework extracts interpretable pose relationship features describing body alignment, joint-angle relationships, body

symmetry, relative body distances, and temporal posture stability. These relationship features are subsequently used for prototype learning, feature space visualization, and prototype-based explainable coaching.

The experiments were designed to evaluate the feasibility of the proposed pose relationship representation, prototype learning, and prototype-based explainable coaching.

4.2 Relationship Feature Analysis

The proposed pose relationship representation is organized into five complementary categories because each category captures a complementary aspect of posture geometry and body alignment. Joint angle features describe local limb configurations, body alignment features characterize global trunk and body orientation, relative body distance features quantify spatial relationships between body parts, and body symmetry features measure left-right balance. For each body relationship feature, the mean represents the representative posture geometry across the motion clip, while the standard deviation characterizes temporal posture stability. In the current implementation, prototype learning is performed using the mean relationship features, which represent stable posture geometry, while the standard deviation features are retained to characterize temporal posture stability.

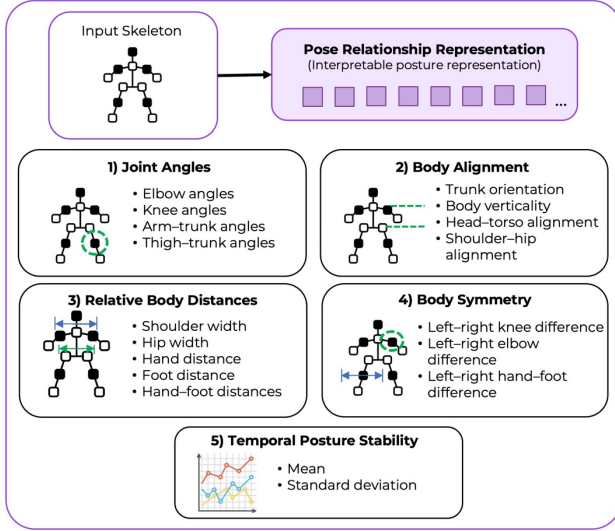


Figure 3. Representative categories of the proposed body relationship representation extracted from an input skeleton sequence

Figure 3 illustrates the representative categories of the proposed pose relationship representation. As shown, an input skeleton is transformed into an interpretable representation composed of joint angles, body alignment, relative body distances, body symmetry, and temporal posture stability. These complementary relationship categories describe posture geometry and alignment in a human-understandable manner, providing an interpretable representation for subsequent prototype learning and posture assessment.

4.3 Visualization of the Learned Prototype Space

To examine whether the proposed pose relationship representations exhibit meaningful structure, a two-dimensional t-SNE visualization was generated from the extracted relationship features. Figure 4 provides a qualitative visualization of the learned relationship feature space and is not used during prototype learning or prototype retrieval. Prototype learning is performed by applying K-Means clustering to the standardized 33-dimensional body relationship feature space, while the t-SNE projection is used solely to qualitatively visualize whether similar posture representations form distinguishable groups.

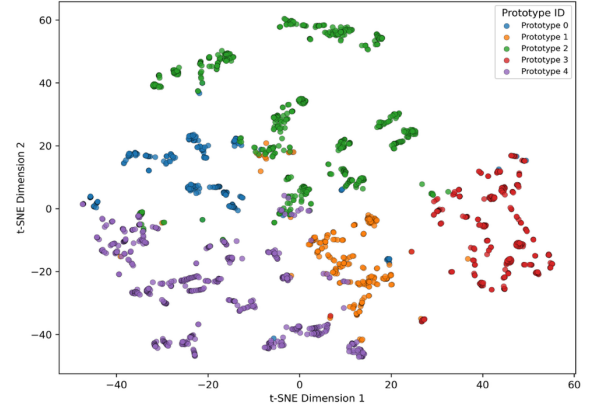


Figure 4. t-SNE visualization of the learned body relationship feature space. Each point represents a posture clip and is colored according to its assigned prototype concept.

As shown in Figure 4, posture representations assigned to the same prototype tend to form localized regions in the embedding space. Although the visualization is not used during prototype learning, the observed grouping qualitatively supports that the proposed body relationship representation preserves meaningful posture similarity, supporting the effectiveness of K-Means clustering for learning representative posture prototypes.

4.4 Explainable Feedback Example

Figure 5 illustrates the complete prototype-based coaching process using the input posture Y007_00150 as an example. The framework first predicts the posture as Upward Facing Dog with a confidence score of 66.33%. Based on the extracted 33-dimensional body relationship representation, the input posture is assigned to Concept 1, whose prototype serves as the most representative body alignment reference for posture assessment.

The retrieved prototype is subsequently compared with the input posture to identify body alignment differences. In this example, the framework detects that the left knee angle and left thigh-trunk angle are lower than the representative prototype, whereas the right thigh-trunk angle and left-right knee angle symmetry are higher. In addition, the distance between the left hand and left foot is smaller than the representative prototype. These prototype-based posture differences provide interpretable evidence describing how the observed posture deviates from the learned body alignment pattern.

The identified posture differences are then converted into a structured prompt containing the predicted pose, interpreted

posture issues, and task instructions before being provided to the Small Language Model (SLM). Based on these structured posture descriptions, the SLM generates concise and personalized coaching feedback without relying directly on raw skeleton coordinates or numerical feature values. Figure 5 demonstrates how prototype-based posture assessment is transformed into human-understandable coaching recommendations.

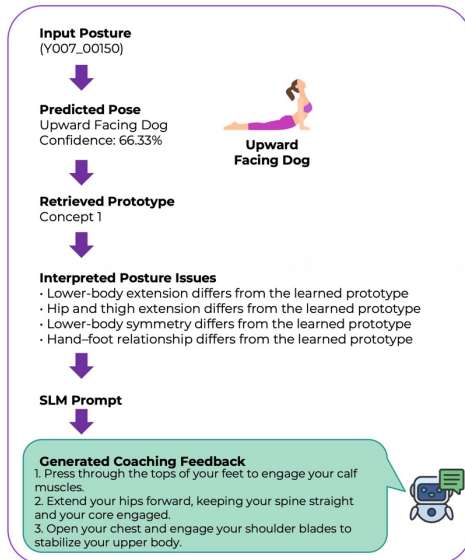


Figure 5. Example of the prototype-based coaching process.

This example illustrates how ProtoCoach-Yoga integrates body relationship representation, prototype-based posture assessment, and SLM-based feedback generation into a transparent and explainable coaching pipeline.

Overall, these preliminary findings demonstrate that interpretable pose relationship representations can be organized into representative prototypes for prototype-based posture assessment. The presented coaching example further shows that the retrieved prototype and identified body alignment differences can be transformed into structured SLM input to generate personalized coaching feedback. Although the current evaluation is primarily qualitative, the results support the feasibility of the proposed ProtoCoach-Yoga framework and provide a foundation for future quantitative evaluation and interactive explainable yoga coaching systems.

5. Conclusion

This paper presented ProtoCoach-Yoga, a positioning framework for explainable yoga coaching that integrates pose relationship prototype learning with Small Language Model (SLM)-based feedback generation. By introducing interpretable pose relationship representations and representative pose relationship prototypes, the proposed framework establishes an intermediate reasoning layer between skeleton-based motion analysis and human-understandable coaching feedback. The learned prototypes provide representative body alignment references for prototype-based posture assessment, enabling explicit body alignment differences to be identified and transformed into structured input for SLM-based coaching

feedback generation. Preliminary qualitative results demonstrate the feasibility of organizing skeleton-based posture information into interpretable representations that support transparent posture assessment and personalized coaching feedback.

Although this work focuses on yoga coaching, the proposed framework has the potential to be extended to a wide range of movement-learning applications, including physical rehabilitation, elderly exercise support, sports coaching, and home fitness guidance. Since the current study presents a positioning framework supported by preliminary qualitative evaluation, future work will focus on comprehensive quantitative evaluation, validation with larger and more diverse datasets, optimization of prototype learning strategies, and user studies to assess the effectiveness, interpretability, and practical usability of the proposed coaching framework.

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